Experiment 9

Text Preprocessing and Feature Scaling Using Python

Subject: Data Science Lab | Dataset: 5 sample sentences | Mapped CO(s): CO2

1. Aim

To preprocess text, convert it to numerical features with TF-IDF, and scale those features with normalization and standardization.

2. Key Concepts

Text preprocessing

Cleaning raw text before a model sees it: lowercase, punctuation removal, tokenization, stop-word removal, then stemming or lemmatization.

Tokenization

Splitting a sentence into words.

Stop words

Common words such as the, and, and of. They are removed because they rarely help a model.

Stemming

Chopping a word to a root, which may not be a real word. Porter stemmer: removes -> remov, characters -> charact.

Lemmatization

Reducing a word to its dictionary form. removes -> remove, characters -> character.

Bag of Words

A table of word counts. CountVectorizer builds it.

TF-IDF

A weight that is high when a word is frequent in one document and rare in the others.

Normalization

Min-Max scaling. Every feature is mapped to the range 0 to 1.

Standardization

Z-score scaling. Every feature is shifted to mean 0 and standard deviation 1.

3. Formulae

TF-IDF (scikit-learn)

$$\text{tfidf}(t, d) = \text{tf}(t, d) \times \text{idf}(t)$$

$$\text{idf}(t) = \ln \frac{1 + n}{1 + \text{df}(t)} + 1$$

n is the number of documents. df(t) is the number of documents that contain term t. Each document vector is then divided by its length (L2 norm).

Min-Max normalization

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Standardization

$$z = \frac{x - \mu}{\sigma}$$

4. What was done

- Loaded 5 sentences about text processing and machine learning.
- Lowercased the text, removed punctuation, tokenized, and removed 198 English stop words.
- Stemmed the remaining tokens with the Porter stemmer.
- Built a TF-IDF matrix of shape 5 documents by 29 features.
- Applied Min-Max scaling and standardization to that matrix.
- Repeated feature extraction with CountVectorizer and compared it with TF-IDF.
- Lemmatized the same tokens and compared them with the stems.
- Scaled a second table (Age, Income, Score) with both scalers.

5. Exercise Answers

Exercise 1: CountVectorizer

CountVectorizer on the stemmed sentences produced a 5 by 29 matrix of raw word counts. Every selected word in these sentences occurred once, so the displayed counts are 0 or 1.

Exercise 2: TF-IDF versus CountVectorizer

Counts only say whether a word is present. TF-IDF gives a smaller weight to a word that appears in more than one document. In document 1, data has TF-IDF 0.379 while learn has 0.306, because data is the rarer of the two in this collection.

Exercise 3: Lemmatization instead of stemming

Word	Stem	Lemma
removes	remov	remove
characters	charact	character
normalize	normal	normalize
learning	learn	learn
documents	document	document

Stemming is faster and cruder. Lemmatization keeps a real word, so the features are easier to read.

Exercise 4: Min-Max on another numerical dataset

Age, Income, and Score were scaled separately. Income values such as 20000 and 120000 became values between 0 and 1, on the same scale as Age and Score.

Exercise 5: Normalization versus standardization

After Min-Max, each column has minimum 0 and maximum 1. After standardization, each column has mean 0 and standard deviation 1. Min-Max is used when a bounded range is needed. Standardization is used when a model assumes features are centered, as many distance-based models do.

6. Viva Questions

Q: Why preprocess text?

A: Raw text has capitals, punctuation, and words like the that do not help the model.

Q: Stemming versus lemmatization?

A: Stemming chops the word. Lemmatization returns the dictionary form.

Q: What does TF-IDF measure?

A: How important a word is in one document compared with the whole collection.

Q: Why scale features?

A: A column like Income would otherwise dominate a column like Age.

Q: Min-Max versus standardization?

A: Min-Max maps values to 0-1. Standardization maps them to mean 0 and standard deviation 1.

Q: What did CountVectorizer produce?

A: A bag-of-words count matrix for the same five sentences.